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# ASTOUNDING Science-Fiction

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## THE WORLD IS MINE

*By Lewis Padgett*

JUNE • 1943

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A STREET AND SMITH PUBLICATION


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# Calling The Empress

## By George O. Smith

**Illustrated by Williams**

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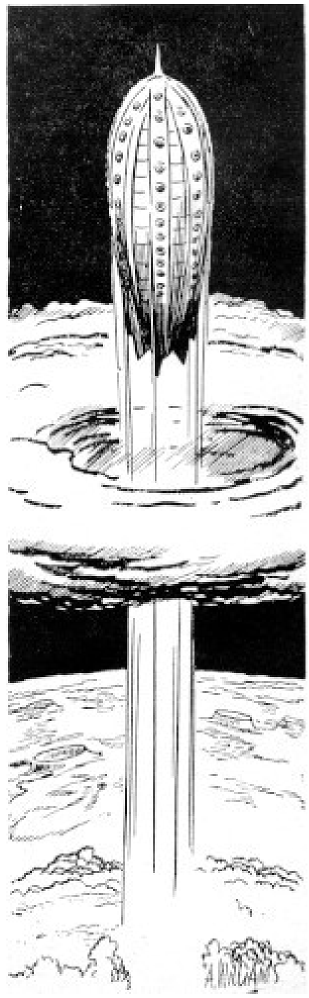
The chart in the terminal building at Canalopsis Spaceport, Mars, was a huge thing that was the focus of all eyes. It occupied a thirty-by-thirty space in the center of one wall, and it had a far-flung iron railing about it to keep the people from crowding it too close, thus shutting off the view. It was a popular display, for it helped to drive home the fact that space travel was different from anything else. People were aware that their lives had been built upon going from one fixed place to another place, equally immobile. But in Interplanet travel one left a moving planet for another planet, moving at a different velocity. You found that the shortest distance was not a straight line but a space curve involving higher mathematics.

The courses being traveled at the time were marked, and those that would be traversed in the very near future were drawn upon the chart, too, all appropriately labeled. At a glance, one could see that in fifty minutes and seventeen seconds, the *Empress of Kolain* would take off from Mars, which was the red disk on the right; and she would travel along the curve so marked to Venus, which was almost one hundred and sixty degrees clockwise around the Sun. People were glad of the chance to go on this trip because the famous Relay Station would come within a telescope's sight on the way.

The *Empress of Kolain* would slide into Venus on the day side and a few hours later she would lift again to head for Terra, a few degrees ahead of Venus and about thirty million miles away.

Precisely on the zero-zero, the *Empress of Kolain* lifted upward on four tenuous pillars of dull-red glow and drove a hole in the sky. The glow was almost lost in the bright sunshine, and soon it died. The *Empress of Kolain* was a little world in itself, and would so remain until it dropped onto the ground at Venus, almost two hundred million miles away.



Driving upward, the *Empress of Kolain* could not have been out of the thin Martian atmosphere when a warning bell rang in the telephone and telespace office at the terminal. The bell caught official ears, and all work was stopped as the personnel of the communications office ran to the machine to see what was so important that the "immediate attention" signal was rung.

Impatiently the operator waited for the tape to come clicking from the machine. It came, letter by letter, click by click, at fifty words per minute. The operator tore the strip from the machine and read aloud: "Hold *Empress of Kolain*. Reroute to Terra direct. Will be quarantined at Venus. Whole planet in epidemic of Venusian Fever."

"Snap answer," growled Keg Johnson. "Tell 'em: 'Too little and too late. *Empress of Kolain* left thirty seconds before warning bell. What do we do now?'"

The operator's fingers clicked madly over the keyboard. Across space went the reply, across the void to the Relay Station. It ran through the Station's mechanism and went darting to Terra. It clicked out as sent in the offices of Interplanet Transport. A vice president read the message and swore roundly. He swore in three Terran languages, in the language of the Venusians, and even managed to visualize a few choice remarks from the Martian Pictographs that were engraved on the Temples of Canalopsis.

"Miss Deane," he yelled at the top of his voice. "Take a message! Shoot a line to Channing on Venus Equilateral. Tell him: '*Empress of Kolain* on way to Venus. Must be contacted and rerouted to Terra direct. Has million dollars' worth of Martian Line Moss aboard; will perish under quarantine. Spare no expense.' Sign that 'Williams, Interplanet.'"

"Yes, Mr. Williams," said the secretary. "Right away."

More minutes of light-fast communication. Out of Terra to Luna, across space to Venus Equilateral Relay Station, the nerve center of Interplanetary Communications. The machines clicked and tape cleared from the slot. It was pasted neatly on a sheet of official paper, stamped *rush* and put in a pneumatic tube.

As Don Channing began to read the message, Keg Johnson on Mars was chewing worriedly on his fourth fingernail, and Vice President Williams was working on his second. But Johnson had a head start and therefore would finish first. Both men knew that nothing more could be done. If Channing couldn't do it, nobody could.

Channing finished the 'gram and swore. It was a good-natured swear word, far from downright vilification, though it did consign certain items to the Nether Regions. He punched a button with some relish, and a rather good-looking woman entered. She smiled at him with more intimacy than a secretary should, and sat down.

"Arden, call Walt in, will you?"

Arden Wastphal smiled. "You might have done that yourself," she told him. She reached for the call button with her left hand, and the diamond on her fourth finger glinted like a pilot light.

"I know it," he answered, "but that wouldn't give me a chance to see you."

"Baloney," said Arden. "You just wait until next October. I'll be in your hair all the time then."

"By then I may be tired of you," said Channing with a smile. "But until then, take it or leave it." His face grew serious, and he tossed the message across the table to her. "What do you think of that?"

Arden read, and then remarked: "That's a huge order, Don. Think you can do it?"

"It'll cost plenty. I don't know whether we can contact a ship in space. It hasn't been done to date, you know, except for short distances."

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The door opened without a knock and Walt Franks entered. "Billing and cooing?" he asked. "Why do you two need an audience?"

"We don't," answered Don. "This was business."

"For want of evidence, I'll believe that. What's the dope?"

"Walt, what are the chances of hooking up with the *Empress of Kolain*, which is en route from Mars to Venus?"

"About equal to a celluloid snowball—you know where," said Franks, looking slyly at Arden.

"Take off your coat, Walt. We've got a job."

"You mean—Hey! Remind me to quit Saturday."

"This is dead in earnest, Walt." Don told the electronics engineer all he knew.

"Boy, this is a job that I wouldn't want my life to depend on. In the first place, we can't beam a transmitter at them if we can't see 'em. And in the second place, if we did, they couldn't receive us."

"We can get a good idea of where they are and how they're going," said Channing. "That is common knowledge."

"Astronomy is an exact science," chanted Franks. "But by the time we figure out just where the *Empress of Kolain* is with respect to us at any given instant we'll all be old men with gray beards. She's crossing toward us on a skew curve—and we'll have to beam it past Sol. It won't be easy, Don. And then if we do find them, what do we do about it?"

"Let's find them first and then work out a means of contacting them afterward."

"Don," interrupted Arden, "what's so difficult?"

Franks fell backward into a chair. Don turned to the girl and asked: "Are you kidding?"

"No. I'm just ignorant. What is so hard about it. We shoot beams across a couple of hundred million miles like nothing and maintain communications at any cost. What should be so hard about contacting a ship?"

"In the first place, we can see a planet, and they can see us, so they can hold their beams. A spaceship might be able to see us, but they couldn't hold a beam on us because of the side sway. We couldn't see them until they are right upon us and so we could not hope to hold a beam on them. Spaceships *might* broadcast, but you have no idea what the square law of radiated power will do to a broadcast signal when millions upon millions of miles are counted in. A half million watts on any planet will not quite cover the planet as a service area on broadcast frequencies. On short waves it will because of the skip distance. But for square-law dissipation, you can't count skip distances—and in space it would be a case of the signal losing in strength according to the inverse square of the distance. So they don't try it. A spaceship may as well be on Rigel as far as contacting her in space goes.

"We might beam a wide-dispersion affair at them," continued Channing. "But it would be pretty thin by the time it got there. And, having no equipment, they couldn't hear us."

"May we amend that?" asked Franks. "They are equipped with radio. But the things are used only in landing operations where the distance is measured in miles, not Astronomical Units."

"O.K.," smiled Channing. "It's turned off during flight and we may consider the equipment as being nonexistent."

"And, according to the chart, we've got to contact them before the turn-about," offered Arden. "They must have time to deflect their course to Terra."

"You think of the nicest complications," said Channing. "I was just about to hope that we could flash them or grab at 'em with a skeeter. But we can't wait until they pass us."

"That will be the last hope," admitted Franks. "But say! Did any bright soul think of shooting a fast ship after them from Canalopsis?"

"Sure. The answer is the same as Simple Simon's answer to the Pieman: Alas, they haven't any!"

"No use asking why," growled Franks. "O.K., Don, we'll go after 'em. I'll have the crew set up a couple of mass detectors at either end of the station. We'll triangulate, and calculate, and hope to hit the right correction factor. We'll find them and keep them in line. You figure out a means of contacting them, huh?"

"*I'll* set up the detectors and *you* find the means," suggested Don.

"No go. You're the director of communications."

Don sighed a false sigh. "Arden, hand me my electronics text," he said.

"And shall I wipe your fevered brow?" cooed Arden.

"Leave him alone," directed Franks. "You distract him."

"It seems to me that you two are taking this rather lightly," said Arden.

"What do you want us to do? Get down on the floor and chew on the rug? You know us better than that. If we can find the answer to contacting a spaceship in flight, we'll add another flower to our flag. But we can't do it by clawing through the first edition of Henney's 'Handbook of Radio Engineering.' It will be done by the seat of our pants if at all; a pair of side-cutters and a spool of wire, a hunk of string and a lump of solder, a—"

"A rag, a bone, and a hank of hair?" asked Franks.

"Leave Kipling out of this. He didn't have to cover the whole Solar System. So let's get cooking."

Don and Walt left the office just a trifle on the fast side. Arden looked after them, out through the open door, shaking her head until she remembered something that she could do. She smiled and went to her typewriter and pounded out a message back to Williams at Interplanet. It read: "Channing and Franks at work on contacting the *Empress of Kolain*. Will do our best." And she signed it: "Venus Equilateral."

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Unknowing of the storm, the *Empress of Kolain* sped silently through the void, accelerating constantly at one gee. Hour after hour she was adding to her velocity, building it up to a speed that would make the trip in days, and not weeks. Her drivers flared dull red no more, for there was not atmosphere for the electronic stream to excite. Her few portholes sparkled with light, but they were nothing in comparison to the starry curtain of the background.

Her hull was of a neutral color, and though the sun glanced from her metal flanks, a reflection from a convex side is not productive of a beam of light. It spreads according to the degree of convexity and is soon lost.

What constitutes an apparent absence? The answer to that question is the example of a ship in space flight. The *Empress of Kolain* did not radiate anything detectable in the electromagnetic scale from ultralong waves to ultrahigh frequencies; nothing at all that could be detected at any distance beyond a few thousand miles. The sweep of her meteor-spotting equipment would pass a spot in micro-seconds at a hundred miles; at the distance from the Relay Station, the sweep of the beam would be curved like the stream of water from a swung hose and therefore useless for direction finding, even though the Station's excellent equipment could pick up the signal. And so fleeting would be the touch of the spotting beam that the best equipment ever known or made would have no time to react, thus marking the signal.

Theorists claim a thing nonexistent if it cannot be detected. The *Empress of Kolain* was invisible. It was undetectable to radio waves. It was in space, so no physical wave could be transmitted to be depicted as sound. Its mass was inconsiderable. Its size was comparatively sub-microscopic, and therefore it would occult few, if any, stars. Therefore, to all intents and purposes, the *Empress of Kolain* was nonexistent, and would remain in that state of material-non-being until it came to life again upon its landing at Venus.

Yet the *Empress of Kolain* existed in the minds of the men who were to find her. Like the shot unseen, fired from a distant cannon, the *Empress of Kolain* was coming at them with ever-mounting velocity, its unseen course a theoretical curve.

And the ship, like the projectile, would land if the men who knew of her failed in their purpose.

Don Channing and Walt Franks found their man in the combined dining room and bar—the only one in many million miles. They surrounded him, ordered a sandwich and beer, and began to tell him their troubles.

Charles Thomas listened for about three minutes. "Boy," he grinned, "being up in that shiny, plush-lined office has sure done plenty to your think-tank, Don."

Channing stopped talking. "Proceed," he said. "In what way has my perspective been warped."

"You talk like Burbank," said Thomas, mentioning a sore spot of some months past. "You think a mass detector would work at this distance? Nuts, fellow. It might, if there were nothing else in the place to interfere. But you want to shoot out near Mars. Mars is on the other side of the Sun—an Evening Star to anyone on Terra. You want us to shoot a slap-happy beam like the mass detector out past Sol; and then a hundred and forty million miles beyond in the faint hope that you can triangulate upon a little mite of matter; a stinking six hundred-odd feet of aluminum hull mostly filled with air and some machinery and so on. Brother, what do you think all the rest of the planets will do to your little piddling beam? Retract, or perhaps abrogate the law of universal gravitation?"

"Crushed," said Franks with a sorry attempt at a smile.

"*Phew!*" agreed Channing. "Maybe I should know more about mass detectors."

"Forget it," said Charles. "The only thing that mass detectors are any good for is to conjure up beautiful bubble dreams, which anybody who knows about 'em can break with the cold point of icy logic."

"What would you do?" asked Channing.

"Darned if I know. We might flash 'em with a big mirror—if we had a big mirror and they weren't heading right into the Sun."

"Let's see," mused Franks, making tabulations on the tablecloth. "They're a couple of hundred million miles away. In order that your mirror present a recognizable disk, it should be about twice the diameter of Venus as seen from Terra. That's eight thousand miles in—at best visibility—say, eighty million or a thousand-to-one ratio. The *Empress of Kolain* is heading at us from some two hundred million, so at a thousand-to-one ratio our mirror would have to be twenty thousand miles across. Some mirror!"

Don tipped Walt's beer over the edge of the table, and while the other man was busy mopping up and muttering unprintables, Don said to Thomas: "This is serious and it isn't. Nobody's going to lose their skin if we don't, but a problem has been put to us and we're going to crack it if we have to skin our teeth to do it."

"You can't calculate their position?"

"Sure. Within a couple of thousand miles we can. That isn't close enough."

"No, it isn't," agreed Chuck.

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Silence fell for a minute. It was broken by Arden, who came in waving a telegram. She sat down and appropriated Channing's glass, which had not been touched. Don opened the sheet and read: "Have received confirmation of your effort. I repeat, spare no expense!" It was signed: "Williams, Interplanet."

"Does that letter offer mean anything to you?" asked Arden.

"Sure," agreed Don. "But at the time we're stumped. Should we be doing something?"

"Anything, I should think, would be better than what you're doing at present. Or does that dinner-and-beer come under the term 'Expenses'?"

Arden stood up, tossed Channing's napkin at him, and started toward the door. Channing watched her go, his hand making motions on the tablecloth. His eyes fell to the table and he took Franks' pencil and drew a long curve from a spot of gravy on one side of the table to a touch of coffee stain on the other. The curve went through a bit of grape jelly near the first stain.

"Here goes the tablecloth strategist," said Franks. "What now, little man?"

"That spot of gravy," explained Don, "is Mars. The jelly is the *Empress of Kolain*. Coffee stain is Venus, and up here by this cigarette burn is Venus Equilateral. Get me?"

"Yop, that's clear enough."

"Now it would be the job for seventeen astronomers for nine weeks to predict the movement of the jelly spot with respect to the usual astral standards. But, fellows, we know the acceleration of the *Empress of Kolain*, and we know her position with respect to the orbit of Mars at the instant of take-off. We can correct for Mars' advance along her—or his—orbit. We can figure the position of the *Empress of Kolain* from her angular distance from Mars! That's the only thing we need know. We don't give a ten-dollar damn about her true position."

Channing began to write equations on the tablecloth. "You see, they aren't moving so fast with respect to us. The course is foreshortened as they are coming almost in line with Venus Equilateral, curving outward and away from the Sun. Her course, as we see it from the Station here, will be a long radius upward curve, slightly on the parabolic side. Like all long-range cruises, the *Empress of Kolain* will heist herself slightly above the plane of the ecliptic to avoid the swarm of meteors that follow about the Sun in the same plane as the planets, lifting the highest at the point of greatest velocity."

"I get it," said Franks. "We get the best beam controller we have to keep the planet on the cross hairs. We apply a spiral cam to advance the beam along the orbit. Right?"

"Right." Don sketched a conical section on the tablecloth and added dimensions. He checked his dimensions against the long string of equations, and nodded. "We'll drive this cockeyed-looking cam with an isochronic clock, and then squirt a beam out there. Thank the Lord for the way our beam transmitters work."

"You mean the effect of reflected waves," asked Chuck.

"Sure. They're like light—only they ain't. We're going to use a glorified meteor detector. We'll control the spread and dispersion so that we cover a healthy hundred miles or so, which will give us sufficient power, I believe. If not, we'll have to tighten the beam. At any rate, spreading from a point source to an object of a given dimension, the waves rebound as though the object were a plane mirror. That will give us a dispersion of twice the dimension of the *Empress of Kolain's* planar projection through this axis. Twelve hundred feet isn't much, but once we get her on the beam and have confirmation, we can forget the rebound. We'll have her pinned."

"And then?" asked Franks.

"Then we will have left the small end, which I'll give to you, Walt, so that you can have part of the credit."

Walt shook his head. "The easy part," he said uncheerfully. "By which you mean the manner in which we contact them and make them listen to us?"

"That's her," said Don with a cheerful smile.

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"Fine!" said Thomas. "Now what do we do?"

"Clear up this mess so we can make the cam. This drawing will do, just grab the tablecloth."

Joe, the operator of Equilateral's one and only establishment for the benefit of the stomach, came up as the three men began to move their glasses and dishes over to an empty table. "What makes with the tablecloth?" he asked. "Don't you want a piece of carbon paper and another tablecloth?"

"No," said Don nonchalantly. "This single copy will do."

"We lose lots of tablecloths that way," said Joe. "It's tough, running a restaurant on Equilateral. I tried using paper ones once, but that didn't work. I had 'em printed, but when the Solar System was on 'em, you fellows drew schematic diagrams for a new coupler circuit. I put all kinds of radio circuits on them, and the gang drew plans for antenna arrays. I gave up and put pads of paper on each table, and the boys used them to make folded paper airplanes and they shot them all over the place. Why don't you guys grow up?"

"Cheer up, Joe. But if this tablecloth won't run through the blueprint machine, we'll squawk!"

Joe looked downcast, and Franks hurried to explain: "It isn't that bad, Joe. We won't try it. We just want to have these figures so we won't have to run through the math again. We'll return the cloth."

"Yeah," said Joe at their retreating figures. "And for the rest of its usefulness it will be full of curves, drawings, and a complete set of astrogating equations." He shrugged his shoulders and went for a new tablecloth.

Don, Walt, and Charles took their improvised drawing to the machine shop, where they put it in the hands of the master mechanic.

"This thing has a top requirement," Don told him, "Make it as quick as you can."

Master Mechanic Walton took the cloth and said: "You forgot the note. You know, 'Work to dimensions shown, do not scale this drawing.' Lord, Don, this silly-looking cam will take a man about six hours to do. It'll have to be right on the button all over, no tolerance. I'll have to cut it to the 'T' and then lap it smooth with polishing compound. Then what'll you test it on?"

"Sodium light interferometer. Can you do it in four hours?"

"If nothing goes wrong. Brass all right?"

"Anything you say. It'll only be used once. Anything of sufficient hardness to withstand a single usage will do."

"I'll use brass then. Or free-cutting steel may be better. If you make it soft you have the chance of cutting too much off with your lapping compound. We'll take care of it, Don. The rest of this stuff isn't too hard. Your framework and so on can be whittled out and pasted together from standard girders, right?"

"Sure. Plaster them together any way you can. And we don't want them painted. As long as she works, phooey to the looks."

"Fine," said Walton. "I'll have the business installed in the Beam Control Room in nine hours. Complete and ready to work."

"That nine hours is a minimum?"

"Absolutely. After we cut and polish that screwball cam, we'll have to check it, and then you'll have to check it. Then the silly thing will have to be installed and its concentricity must be checked to the last wave length of cadmium light. That'll take us a couple of hours, I bet. The rest of the works will be ready, checked, and waiting for the ding-busted cam."

"Yeah," agreed Franks. "Then we'll have to get up there with our works and put the electricals on the mechanicals. My guess, Don, is a good, healthy twelve hours before we can begin to squirt our signal."

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Twelve hours is not much in the life of a man; it is less in the life of a planet. The Terran standard of gravity is so small that it is expressed in feet per second. But when the two are coupled together as a measure of travel, and the standard Terran gee is applied for twelve hours steady, it builds up to almost three hundred miles per second, and by the end of that twelve hours, six million miles have fled into the past.

Now take a look at Mars. It is a small, red mite in the sky, its diameter some four thousand miles. Sol is eight hundred thousand miles in diameter. Six million miles from Mars, then, can be crudely expressed by visualizing a point eight times the diameter of the Sun away from Mars, and you have the distance that the *Empress of Kolain* had come from Mars.

But the ship was heading in at an angle, and the six million miles did not subtend the above arc. From Venus Equilateral, the position of the *Empress of Kolain* was more like two diameters of the Sun from Mars, slightly to the north and on the side away from Sol.

It may sound like a problem for the distant future, this pointing a radio beam at a planet, but it is no different than Galileo's attempts to see Jupiter through his Optik Glass. Of course, it has had refinements that have enabled men to make several hundred hours of exposure of a star on a photographic plate. So if men can maintain a telescope on a star, night after night, to build up a faint image, they can also maintain a beamed transmission wave on a planet.

All you need is a place to stand; a firm, immobile platform. The three-mile-long, one-mile-diameter mass of Venus Equilateral offered such a platform. It rotated smoothly, and upon its 'business' end a hardened and highly polished set of rails maintained projectors that were pointed at the planets. These were parabolic reflectors that focused ultra-high-frequency waves into tight beams which were hurled at Mars, Terra, and Venus for communication.

And because the beams were acted upon by all of the trivia in the Solar System, highly trained technicians stood their tricks at the beam controls, correcting by sensitive verniers any deviation of the beams. In fifty million miles, even the bending of electromagnetic waves by the Sun's mass had to be considered. Sunspots made known their presence. And the vagaries of land transmission were present in a hundred ways due to the distance and the necessity of concentrating every milliwatt of available power on the target.

This problem of the *Empress of Kolain* was different. Spaceships were invisible, therefore the beam-control man must sight on Mars and the mechanical cam would keep the ship in sight of the beam.

The hours went past in a peculiar mixture of speed and slowness. On one hand the minutes sped by swiftly and fleetingly, each tick of the clock adding to the lost moments, never to be regained. Time, being precious, seemed to slip through their fingers like sifting sand.

On the other hand, the time that must be spent in preparation of the equipment went slow. Always it was in the future, that time when their experiment must either prove a success or a failure. Always there was another hour of preparatory work before the parabolic reflector was mounted; and then another hour before it swung freely and perfectly in its new mounting. Then the minutes were spent in anticipation of the instant that the power stage of the transmitter was tested and the megawatts of ultra-high-frequency energy poured into the single rod that acted as a radiator.

It was a singularly disappointing sight. The rod glowed not, and the reflector was the same as it was before the rod drew power. But the meters read and the generators moaned, and the pyrometers in the insulators mounted as the losses converted the small quantity of energy into heat. But the rod drew power, and the parabolic reflector beamed that power into a tight beam and hurled it out on a die-true line.

Invisible power that could be used in communications.

Then the cam was installed. The time went by even slower then, because the cam must be lapped and polished to absolute perfection, not only of its own surface but to absolute concentricity to the shaft on which it turned.

But eventually the job was finished, and the men stood back, their eyes expectantly upon Don Channing and Walt Franks.

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Don spoke to the man chosen to control the beam. "You can start any time now. Keep her knifed clean, if you can."

The man grinned at Channing. "If the devils that roam the void are with us we'll have no trouble. We should all pray for a phrase used by some characters in a magazine I read once: 'Clear ether!' We could use some right now."

He applied his eyes to the telescope. He fiddled with the verniers for a brief time, made a major adjustment on a larger handwheel, and then said, without removing his eye from the 'scope, "That's it, Dr. Channing."

Don answered: "O.K., Jim, but you can use the screen now. We aren't going to make you squint through that pipe for the next few hours straight."

"That's all right. I'll use the screen as soon as we can prove we're right. Ready?"

"Ready," said Channing.

Franks closed a tiny switch. Below, in the transmitter room, relays clicked and heavy-duty contacts closed with blue fire. Meters began to climb upward across their scales, and the generators moaned in a descending whine. A shielded monitor began to glow, indicating that full power was vomiting from the mouth of the reflector.

And out from the projector there went, like a spearhead, a wavefront of circularly polarized microwaves. Die-true they sped, crossing the void like a line of sight to an invisible spot above Mars and to the left. Out past the Sun, where they bent inward just enough to make Jim's job tough. Out across the open sky they sped at the velocity of light, and taking sixteen minutes to get there.

Would it—or wouldn't it?

A half-hour passed. "Now," said Channing. "Are we?"

Ten minutes went by. The receiver was silent save for a constant crackle of cosmic static.

Fifteen minutes passed.

"Nuts," said Channing. "Could it be that we aren't quite hitting them?"

"Could be," admitted Franks. "Jim, waggle that beam a bit, and slowly. When we hit 'em, we'll know it because we'll hear 'em a half-hour later. Take it easy and slowly. We've used up thirteen of our fifty-odd hours. We can use another thirty or so just in being sure."

Jim began to make the beam roam around the invisible spot in the sky. He swept the beam in microscopic scans, up and down, and advancing the beam by one half of its apparent width at the receiver for each sweep.

Two more hours went by. The receiver was still silent of reflected signals.

It was a terrific strain, this necessary wait of approximately a half-hour between each minor adjustment and the subsequent knowledge of failure. Jim gave up the 'scope because of eyestrain, and though Don and Walt had confidence that the beam-control man was competent to use the cross-ruled screen to keep Mars on the beam, Jim was none too sure of himself, and so he kept checking the screen against the 'scope.

At the end of the next hour of abject failure, Walt Franks began to scribble on a pad of paper. Don came over to peer over Franks' shoulder, and because he couldn't read Walt's mind, he was forced to ask what the engineer was calculating.

"I've been thinking," said Franks.

"Beginner's luck?" asked Don with a wry smile.

"I hope not. Look, Don, we're moving on the orbit of Venus, at Venus' orbital velocity. Oh, all right, say it scientifical: We're all three, Venus, Sol, and Venus Equilateral, at the corners of an equilateral triangle, and will forever remain, barring outside influences. So that means we're running around a common point, the common center of gravity—which can be construed to mean that we are circling Sol at Venus' speed, or twenty-one point seven five miles per second. Now our beam is curved because of the angular velocity, just like a swung hose. However, it hits the *Empress of Kolain* at an angle as though we were a couple of thousand feet away. That's fine. But the reflected wave starts back at that angle, right back through the beam, remember?"

"I get it!" shouted Don in glee. "Thirty-two seconds at twenty-one point seven miles per second gives us seven hundred and sixteen miles to the rear. Walt, get your mechanical gang to hitch us up a couple of mirrors—say a yard in diameter. Put 'em so that they can be used as a range finder. Set the angles for seven hundred and sixteen miles; a three-mile base line should do it, I'm sure; and then we'll shoot us a skeeter out there with a detector. Get carving!"

"Shall Jim stop?" asked Walt.

"How long will it take to rig us a range finder?"

"Hour, God willing."

"Jim, get a relief for a half-hour. We'll keep the beam centered. Then he can take over when the going gets critical again."

---

The mounting of two mirrors at either end of Venus Equilateral gave little trouble. It was the amount of detailed work that consumed the time. There were girders to be cut and welded together. The hundred-odd doorways that centered on the axis of Venus Equilateral had to be opened and the clear, light path had to be cleared of packing cases, supplies, and in a few cases machinery had to be partially dismantled to clear the way. A good portion of Venus Equilateral's personnel of three thousand were taken off of their jobs, haled out of bed for the emergency, or made to work through their play period, depending upon which shift they worked.

The machinery could be replaced, the central storage places could be refilled, and the many doors closed again. But the central room containing the air plant was no small matter. Channing took a sad look at the lush growth of Martian saw grass and sighed. It was growing nicely now, they had nurtured it into lusty growth from mere sprouts in trays and it was as valuable—precisely—as the lives of the three thousand-odd that lived, loved, and pursued happiness on Venus Equilateral. It was a youthful plant, a replacement brought in a tearing hurry from Mars to replace the former plant that was heaved into the incinerator by a well-meaning but ignorant man who thought that an air plant must be huge, moving levers, whirling gears, bubbling retorts, and a sprig of parsley.

Channing closed his eyes and shuddered in mock horror. "Chop out the center," he said.

The "center" meant the topmost fronds of the long blades; their roots were embedded in the trays that filled the cylindrical floor. Some of the blades would die—Martian saw grass is tender in spite of the wicked spines that line the edge—but this was an emergency with a capital E.

Cleaning the centermost channel out of the station was no small job. The men who put up Venus Equilateral had no idea that someone would be using the station for a sighting tube some day. The many additions to the station through the years made the layout as regular and as well-planned as the Mammoth Cave in Kentucky.

So for hour upon hour, men swarmed in the central, weightless channel and wielded acetylene torches, cutting steel. Not in all cases, but there were many. In three miles of storage rooms, a lot of doors and bulkheads can be thrown up without crowding the size of the individual rooms.

Channing spoke into the microphone at the North end of Venus Equilateral, and said: "Walt? We've got a sight. Can you see?"

"Yop," said Walt. "And say, what happens to me after that bum guess?"

"That was quite a stretch, Walt. That 'hour, God willing,' worked itself into four hours, God help us."

"O.K., so I was optimistic. I thought that those doors were all on the center line."

"They are supposed to be, but they aren't huge and a little misalignment can do a lot of light-stopping. Can we juggle mirrors now?"

"Sure as shooting. Is Freddy out in the flitter?"

"He says he is guessing that he is at the right distance now. I'm set at a right angle now"—Don began to fumble through a volume of Vega's "Logarithms and Trigonometrical Functions"—"eighty-nine degrees, forty-five minutes, and forty seconds. Can you set your mirror that close?"

"Nope," answered Walt without a qualm. "Not a chance. I can hit it about ten seconds plus or minus, though."

"Make it plus nothing, minus twenty," said Don. "I was playing by ear, this time on account of my slipstick is busted."

"Such a lot of chatter," returned Walt, "don't mean a thing. While you've been gabbing about your prowess with a busted slide rule I've been setting my glass. You can cook with glass now."

"Brother," groaned Channing, "if I had one of those death rays that the boys were crowing about back in the days before space hopping became anything but a bit of fiction, I'd scorch your ears—or burn 'em off—or blow holes in you—or disintegrate you—depending on what stories you read. I haven't heard such a lousy pun in seventeen years—Hey, Freddy, you're a little close. Run out a couple of miles, huh?—and, Walt, I've heard some doozies."

There was a click in the phones and a cheerful voice chimed in with: "Good morning, fellows? What's with the Great Quest?"

Channing answered, "Hi, Babe. Been snoozing?"

"Sure, as any sensible person would. Have you been up all the time?"

"Yeah. We're still up against the main trouble with telephones—the big trouble, same as back in 1877—our friends have no telephone! You'd be surprised how elusive a spaceship can be in the deep. Sort of a nonexistent, microscopic speck, floating in absolutely nothing. We have a good idea of where they should be, and possibly why and what—but we're really playing with blindfolds, handcuffs, car plugs, mufflers, nose clamps, and tongue-ties. I am reminded—Hey, Freddie, lift her north about three hundred yards—of the two blind men."

"Never mind the blind men," came back the pilot. "How'm I doing?"

"Fine. Slide out another hundred yards and hold her there."

"Who—me? Listen, Dr. Channing, you're the bird on the tapeline. You have no idea just how insignificant you look from seven hundred and sixteen miles away. Put a red-hot on the 'finder and have 'im tell me where the ship sits!"

"O.K., Freddy, you're on the beam and I'll put a guy on here to give you the dope. Right?"

"Right!"

"Right," echoed Arden. "And I'm going to bring you a slug of coffee and a roll. Or did you remember to eat recently?"

"We didn't," chimed in Walt.

"You get your own girl," snorted Channing. "And besides, you are needed up here. We've got work to do."

---

Once again the signal lashed out. The invisible waves drove out and began their swift run across the void. Time, as it always did during the waiting periods, hung like a Sword of Damocles. The half-hour finally ticked away, and Freddy called in: "No dice. She's as silent as the grave."

Minutes added together into an hour. The concentric wave left the reflector and just dropped out of sight.

"Too bad you can't widen her out," suggested Don.

"I'd like to tighten it down," objected Walt. "I think we're losing power and we can't increase the power—but we could tighten the beam."

"Too bad you can't wave it back and forth like a fireman squirting water on a lawn," said Arden.

"Firemen don't water lawns—" began Walt Franks, but he was interrupted by a wild yell from Channing.

"Something hurt?" asked Arden.

"No. Walt, we can wave the beam."

"Until we find 'em? We've been trying that. No worky."

Freddy called in excitedly: "Something went by just now and I don't think it was Christmas!"

"We might have hit 'em a dozen times in the last ten minutes and we'll never know it," said Channing. "But the spaceliners can be caught. Let's shoot at them like popping ducks. Shotgun effect. Look, Walt, we can electronically dance the beam at a high rate of speed, spraying the neighborhood. Freddy can hear us return because we have to hit them all the time and the waver coming on the way back will pass through his position again and again. We'll set up director elements in the reflector, distorting the electrical surface of the parabolic reflector. That'll divert the beam. By making the phases swing right, we can scan the vicinity of the *Empress of Kolain* like a flying-spot television camera."

Walt turned to one of the technicians and explained. The man nodded. He left for Franks' laboratory and Walt turned back to his friends.

"Here shoots another couple of hours. I, for one, am going to grab forty winks."

Jim, the beam-control man, sat down and lighted a cigarette. Freddy let his flitter coast free. And the generators that fed the powerful transmitter came whining to a stop. But there was no sleep for Don and Walt. They kept awake to supervise the work, and to help in hooking up the phase-splitting circuit that would throw out-of-phase radio frequency into the director-elements to swing the beam.

Then once again the circuits were set up. Freddy found the position again and began to hold it. The concentric beam hurled out again, and as the phase-shift passed from element to element, the beam swept through an infinitesimal arc that covered thousands of miles of space by the time the beam reached the position occupied by the *Empress of Kolain*.

Like a painter, the beam painted in a swipe a few hundred miles wide and swept back and forth, each sweep progressing ahead of the stripe before by less than its width. It reached the end of its arbitrary wall and swept back to the beginning again, covering space as before. Here was no slow, irregular swing of mechanical reflector, this was the electronically controlled wavering of a stable antenna.

And this time the half-hour passed slowly but not uneventfully. Right on the tick of the instant, Freddy called back: "Got 'em."

It was a weakling beam that came back in staccato surges. A fading, wavering, spotty signal that threatened to lie down on the job and sleep. It came and it went, often gone for seconds and never strong for so much as an instant. It vied, and almost lost completely, with the constant crackle of cosmic static. It fought with the energies of the Sun's corona and was more than once the underdog. Had this returning beam carried intelligence of any sort it would have been wasted. About all that could be carried on a beam as sorry as this was the knowledge that there was a transmitter—and that it was transmitting.

But its raucous note synchronized with the paint-brush wiping of the transmitter. There was no doubt.

Don Channing put an arm around Arden's waist and grinned at Walt Franks. "Go to work, genius. I've got the *Empress of Kolain* on the pipe. You're the bright-eyed lad that is going to wake 'em up! We've shot almost twenty hours of our allotted fifty. Make with the megacycles, Walter. Arden and I will take in a steak, a moom pitcher, and maybe a bit of woo. Like?" he asked the girl.

"I like," she answered.

---

Walt Franks smiled and stretched lazily. He made no move to the transmitter. "Don't go away," he cautioned them. "Better call up Joe and order beer and sandwiches for the boys in the back room. On you!"

"Make with the signals first," said Channing. "And lay off the potables until we finish this silly job."

"You've got it. Is there a common, garden variety, transmitting key in the place?"

"Probably. We'll have to ask. Why?"

"Ask me."

Don removed his arm from Arden's waist. He picked up a spanner and advanced upon Franks.

"No!" objected Arden. "Poison him—I can't stand the sight of blood. Or better, bamboo splinters under the fingernails. He knows something simple, the big bum!"

"Beer and sandwiches?" asked Walt.

"Beer and sandwiches," agreed Don. "Now, Tom Swift, what gives?"

"I want to key the inner component of the beam. Y' see, Don, we're using the same frequency, by a half dozen megacycles, as their meteor spotter. I'm going to retune the inner beam to their frequency and key it. Realize what'll happen?"

"Sure," agreed Don, "but you're still missing the boat. You can't transmit keyed intelligence with an intermittent contact."

"In words, what do you mean, Don?"

"International Code is a series of dots and dashes, you may know. Our wabbling beam is whipping through the area in which the *Empress of Kolain* is passing. Therefore the contact is intermittent. And how could you tell a dot from a dash?"

"Easy," bragged Walt Franks. "We're not limited as to the speed of deviation, are we?"

"Yes—limited by the speed of the selsyn motors that transfer the phase-shifting circuits to the director radiators. Yeah, I get it, Edison, and we can wind them up to a happy six or eight thousand r.p.m. Six would get us a hundred cycles per second—a nice, low growl."

"And how will they receive that kind of signal on the meteor spotter?" asked Arden.

"The Officer of the Day will be treated to the first meteor on record that has intermittent duration—it is there only when it spells in International Code!"

---

Prying the toy transmitting key from young James Burke was a job only surpassed in difficulty by the task of opening the vault of the Interplanetary Bank after working hours. But Burke, Junior, was plied with soda pop, ice cream and candy. He was threatened, cajoled, and finally bribed. And what Interplanetary Communications paid for the toy finally would have made the manufacturer go out and look for another job. But Walt Franks carried the key to the scene of operations and set it on the bench to look at it critically.

"A puny gadget, at that," he said, clicking the key. "Might key a couple of hundred watts with it—but not too long. She'd go up like a skyrocket under our load!"

Walt opened a cabinet and began to pull out parts. He piled several parts on a bread board, and in an hour had a very husky thyraton hooked into a circuit that was simplicity itself. He hooked the thyraton into the main power circuit and tapped the key gingerly. The transmitter followed the keyed thyraton and Don took a deep breath.

"Do you know code?" he asked.

"Used to. Forgot it when I came to Venus Equilateral. Used to hold a ham ticket on Terra. But there's no use in hamming on the station here where you can work somebody by yelling at the top of your voice. The thing to ask is, 'Does anybody know code on board the *Empress of Kolain*?'"

They forgot their keying circuit and began to adjust the transmitter to the frequency used by the meteor spotter. It was a job. But it was done, all the way from the master oscillator stage through the several frequency doubler stages and to the big power-driver stage. The output stage came next, and then a full three hours of tinkering with files and hacksaws were required to adjust the length of the main radiator and the director-elements so that their length became right for the changed frequency.

Finally Walt took the key and said: "Here goes!"

He began to rattle the key. In the power room, the generators screamed and the lights throughout the station flickered just a bit at the sudden surges.

Don Channing said to Arden: "If someone on the *Empress of Kolain* can understand code—"

---

The *Empress of Kolain* was zipping along in its silent passage through the void. It was an unseen, undetected, unaware bit of human manufacture marking man's will among the stars. In all the known universe it moved against the forces of celestial mechanics because some intelligent mote that infested the surface of a planet once had a longing to visit the stars. In all the Solar System, most of the cosmic stuff was larger than it—but it alone defied the natural laws of space.

Because it alone possessed the required *outside force* spoken of in Newton's "Universal Laws."

And it was doing fine.

Dinner was being served in the dining room. A group of shapely girls added grace to the swimming pool on the promenade deck. The bar was filled with a merry crowd which in turn were partly filled with liquor. A man in uniform, the Second Officer, was throwing darts with a few passengers in the playroom, and there were four oldish ladies on sabbatical leave who were stricken with *mal-de-void*.

The passage up to now had been uneventful. A meteor or two had come to make the ship swing a bit—but the swerve was less than the pitch of an ocean vessel in a moderate sea and it did not continue as did an ocean ship. Most of the time the *Empress of Kolain* seemed as steady as solid rock.

Only the First Officer, on the bridge, and the Chief Pilot, far below in the control room, knew just how erratic their course truly was. But they were not worried. They were not a shell, fired from a gun; they were a spaceship, capable of steering themselves into any port on Venus when they arrived and the minute wobbulations in their course could be corrected when the time came. For nothing that had come across the universe yet had ever prevented the ship of space from seeing where it was going.

Yes, it was uneventful.

Then the meteor screen flashed into life. A circle of light appeared in the celestial dome and the ship automatic pilot swerved ever so little. The dot of light was gone.

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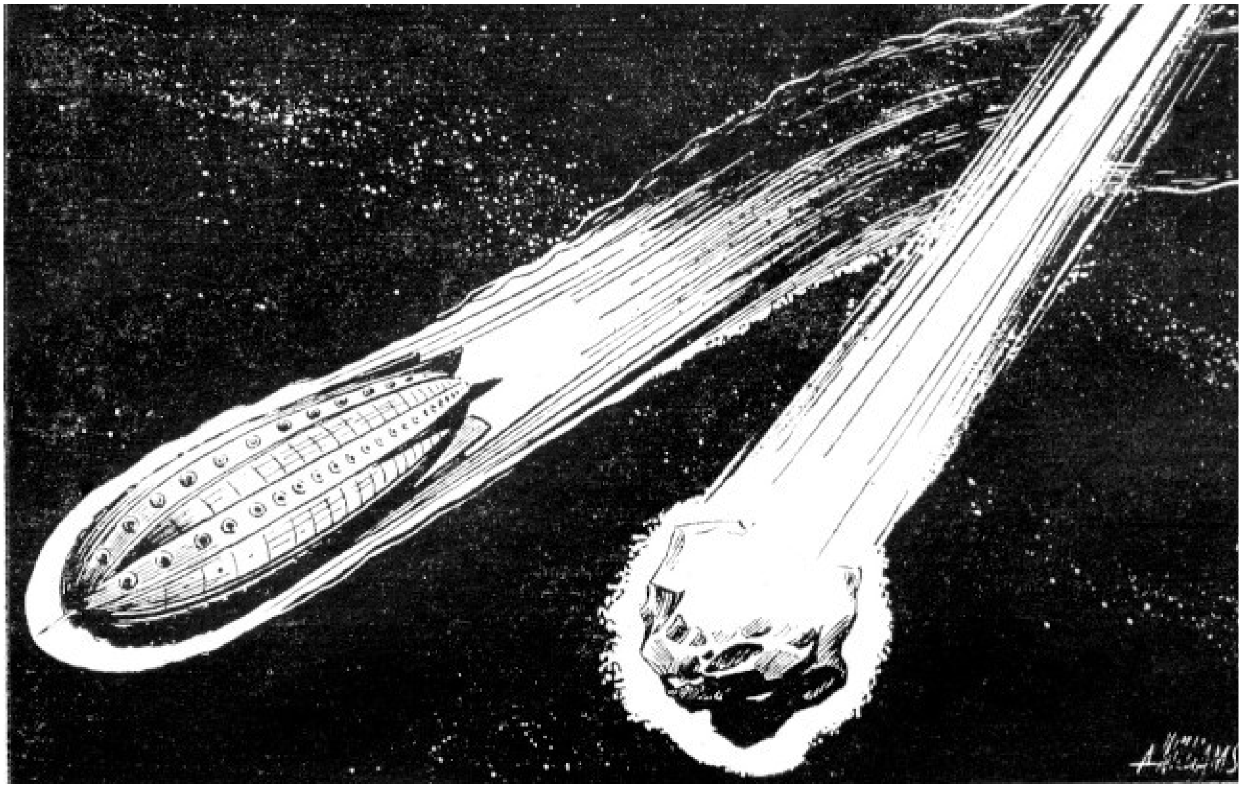

Throughout the ship, people laughed nervously. A waiter replaced a glass of water that had been set too close to the edge of the table and a manly-looking fellow dived into the swimming pool to haul a good-looking blonde back to the edge again. She'd been in the middle of a swan dive when the swerve came and the ship had swerved without her. The resounding smack of feminine stomach against the water was of greater importance than the meteor, now so many hundred miles behind.

The flash of light returned and the ship swerved again. Upon the third swerve, the First Officer was watching the celestial globe with suspicion. He went white. It was conceivable that the *Empress of Kolain* was about to encounter a meteor shower.

And that was bad.

He marked the place and set his observation telescope in synchronism with the celestial globe. There was nothing but the ultimate starry curtain in the background. He snapped a switch and the voice of the pilot came out of a speaker in the wall.

"You called, Mr. Hendall?"

"Tony, take the levers, will you please? Something is rotten in the State of Denmark."

"O.K., sir. I'm riding personal."

"Kick out the meteor-spotter coupling circuits and forget the alarm."

"Right, Mr. Hendall, but will you confirm that in writing?"

Hendall scribbled on the telautograph and then abandoned the 'scope. The flashing in the celestial globe continued, but the ship no longer danced in its path.

The big twenty-inch Cassegrain showed nothing at all, and Hendall returned to the bridge scratching his head. Nothing on the spotting 'scope and nothing on the big instrument.

That intermittent spot was large enough to mean a huge meteor. But wait. At their speed, it should have retrogressed in the celestial globe unless it was so huge and so far away—but Sol didn't appear on the globe and it was big and far away, bigger by far. Nothing short of a planet at less-than-planetary distances would do this.

Not even a visible change in the position of the spot.

"Therefore," thought Hendall, "this is no astral body that makes this spot!"

---

Hendall went to a cabinet and withdrew a cable with a plug on either end. He plugged one end into the test plug on the meteor spotter and the opposite end into a speaker. A low humming emanated from the speaker in synchronism with the flashing of the celestial globe.

It hit a responsive chord.

Hendall went to the main communication microphone and spoke. His voice went all over the *Empress of Kolain* from pilot room and cargo spaces to swimming pool and infirmary.

"Attention!" he said in a formal tone. "Attention to official orders!"

Dancers stopped in midstep. Swimmers paused and then made their way to the edges of the pool and sat with their feet dangling in the warm water. Diners sat with their forks poised foolishly.

"Official orders!" That meant an emergency.

Hendall continued: "I believe that something never before tried is being attempted. I am forced against my better knowledge to believe that some agency is trying to make contact with us; a spaceship in flight! This is unknown in the annals of space flying and is, therefore, indicative of something important. It would not have been tried without preparations unless an emergency exists.

"However, the requirements of an officer of space do not include a knowledge of code because of the lack of communication with the planets while in space. Therefore, I request that any person with a working knowledge of International Morse will please present himself to the nearest officer."

Minutes passed. Minutes during which the flashing lights continued.

Then the door of the bridge opened and Third Officer Jones entered with a thirteen-year-old boy at his heels. The youngster's eyes went wide at the sight of the instruments on the bridge, and he looked around in amazed interest.

"This is Freddy Thomas," said Jones. "He knows code!"

"Go to it, Mr. Thomas," said Hendall.

The boy swelled visibly. You could almost hear him thinking: "He called me 'mister'!"

Then he went to the table by the speaker and reached for pencil and paper. "It's code all right," he said. Then Freddy winked at Jones. "He has a lousy fist!"

Freddy Thomas began to write.

> "—course and head for Terra direct"—the beam faded for seconds—"Venusian fever and you will be quarantined.

> "Calling CQ, calling CQ, calling CQ. Calling *Empress of Kolain* ... empowered us to contact you and convey ... message—You are requested to correct your course and head ... a plague of Venusian fever and you—Williams of Interplanet has empowered us ... the following message: 'You are requested to correct your ... head for Terra direct.' Calling CQ...."

"Does that hash make sense to you?" asked Jones of Hendall.

"Sure," smiled Hendall, "it is fairly plain. It tells us that Williams of Interplanet wants us to head for Terra direct because of a plague of Venusian fever that would cause us to stay in quarantine. That would ruin the Line Moss. Prepare to change course, Mr. Jones!"

"Who could it be?" asked Jones foolishly.

"There is only one outfit in the Solar System that could possibly think of a stunt like this. And that is Channing and Franks of Interplanetary Communications. This signal came from Venus Equilateral."

"Wait a minute," said Freddy Thomas. "Here's some more."

> "'As soon as this signal—intelligible—at right angles to your course for ten minutes. That will take—out of—beam and reflected—will indicate to us—left the area and know of our attempt.'"

"They're using a beam of some sort that indicates to them that we are on the other end but we can't answer. Mr. Jones, and Pilot Canton, ninety degrees north for ten minutes! Call the navigation officer to correct our course. I'll make the announcement to the passengers. Mr. Thomas, you are given the freedom of the bridge for the rest of the trip."

Mr. Thomas was overwhelmed. He'd learn plenty—and that would help him when he applied for training as a space officer; unless he decided to take a position with Interplanetary Communications when he grew up.

---

The signal faded from the little cruiser and silence prevailed. Don spoke into the microphone and said: "Run her up a millisecond," to the beam controller.

The beam wiped the space above the previous course for several minutes and Franks was sending furiously:

"You have answered our message. We'll be seeing you."

Channing told the man in the cruiser to return. He kicked the main switch and the generators whined down the scale and coasted to a stop. Tube filaments darkened and meters returned to zero.

"O.K., Walton. Let the spinach lay. Get the next crew to clean up the mess and polish the set-up into something presentable. I'll bet a cooky that we'll be chasing spaceships all the way to Pluto after this. We'll work it into a fine thing and perfect our technique. Right now I owe the gang a dinner, right?"

---

Nothing ever happens at Venus Equilateral. The weather is always right. It never rains or storms. There is no icy street nor heat-waved plain. There is no mud. There is no summer, no winter, no spring, no fall. People ice skate and swim in adjacent rooms. There is no moon to enchant for the moon is millions of miles away. There is no night or day and the stars blaze out in the same sky with the sun; and it has been said that on Venus Equilateral you have been in the only place where the Clouds of Magellan and Polaris can be seen at the same time from your living-room window.

Venus Equilateral is devoted to the business of supplying communication between the three inner planets. As such, it is more than worth it. And though electromagnetic waves travel with the speed of light in vacuo, Channing and his crew were fast asleep by the time that Williams, of Interplanet, read the following message:

> *EMPRESS OF KOLAIN* CONTACTED AND MESSAGE CONVEYED. SHIP WILL PUT IN AT TERRA AS PER YOUR REQUEST. YOURS FOR BETTER COMMUNICATIONS.
>
> DON CHANNING.

THE END

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